Adjudication summary

Orthogonal confidence layer — SQANTI3 features are consumed as priors, never recomputed.

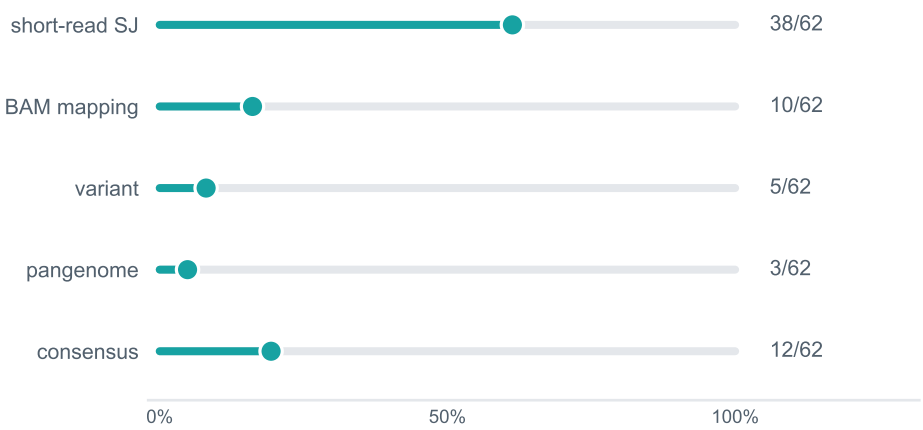
Verdict on 49 novel isoforms

Trustworthy = high/medium-confidence novel · the share you can act on at a glance



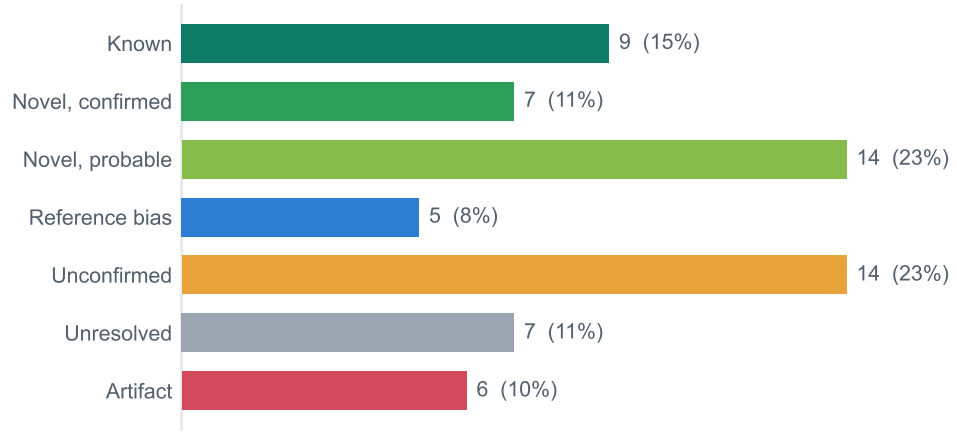
Evidence axes in play

fraction of isoforms for which each axis could be evaluated



Confidence-class distribution

every isoform · count (share of all)

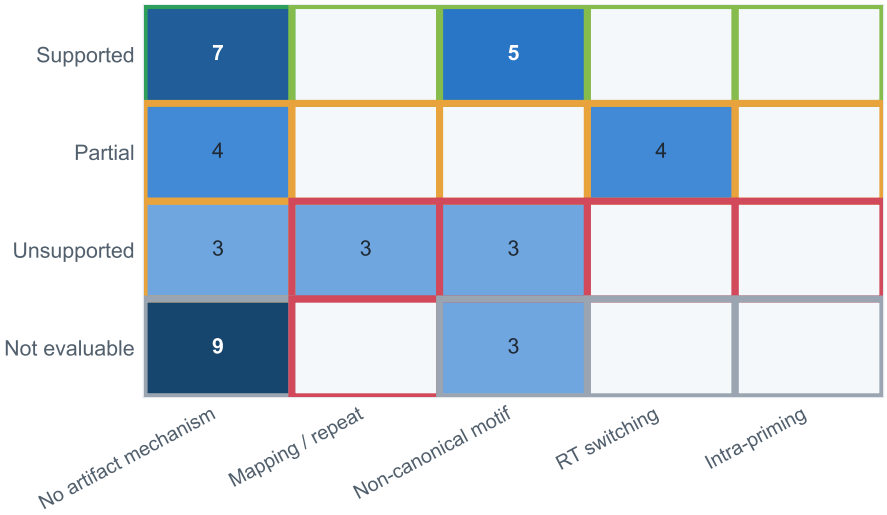


How each verdict was reached

Every novel isoform is scored on two axes; the pair projects to one confidence class.

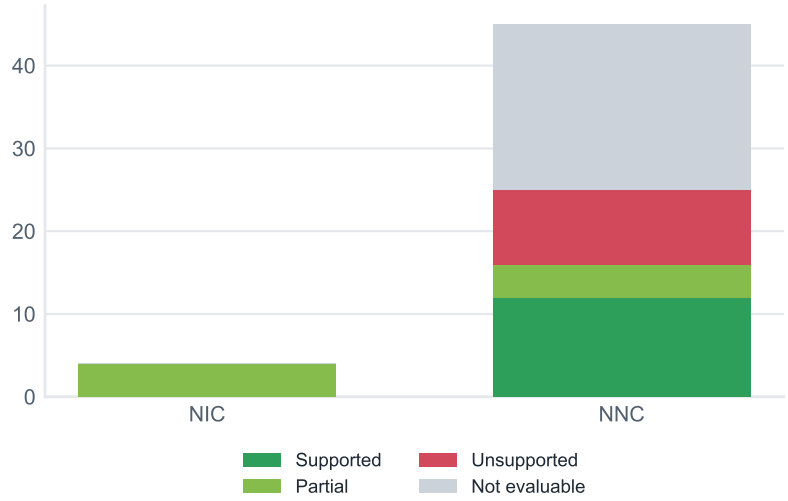
Novelty support × artifact mechanism

cell = isoform count · coloured border = the class this pair projects to
(UNKNOWN row: UNRESOLVED unless caller consensus or ≥ 3 whole-chain reads lift it)



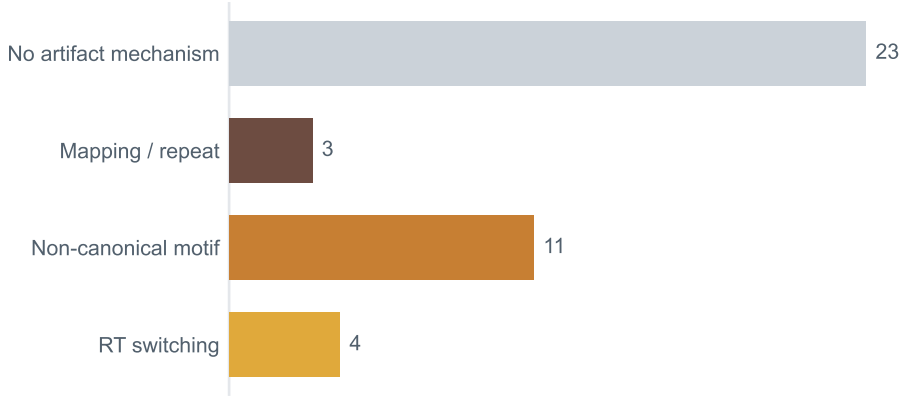
Novelty support by category

is novel-junction support better for NIC or NNC?



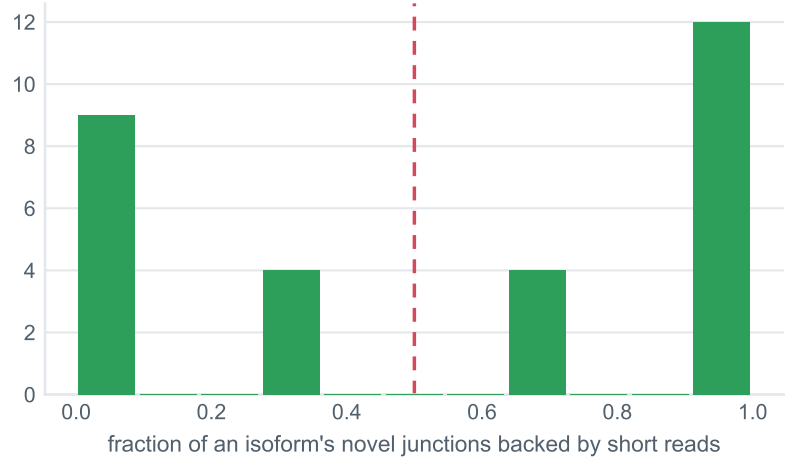
Dominant artifact mechanism

engine priority: mapping > non-canonical > RT-switch > intra-priming



Short-read corroboration

to the right of the dashed line = majority backed



Why calls were down-weighted

Which artifact mechanism drove each verdict, and whether the evidence supports it.

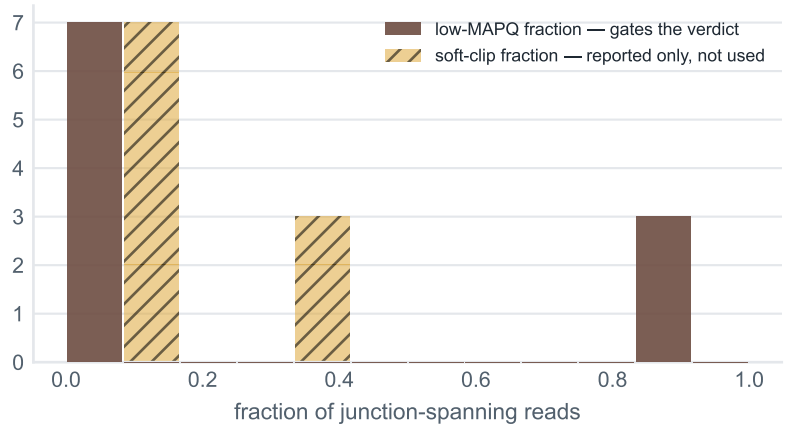
Mechanism → resulting class

where each artifact mechanism ends up



BAM mapping-evidence quality

are mapping artifacts backed by poor alignment?



Structural-signal availability

null is shown as n/a — never silently counted as false

Structural prior	true	false	n/a
Non-canonical motif	11	51	0
RT-switch stage	4	58	0
Caller chain available	62	0	0

Mapping-down-weighted calls (3)

isoform	class	reads	low-MAPQ	suppl.
NNC_map.36	Artifact	14	0.86	0.71
NNC_map.37	Artifact	14	0.86	0.71
NNC_map.38	Artifact	14	0.86	0.71

Reference-bias rescue & circularity firewall

A novel junction explained by reference bias is reclassified — but only on independent evidence.

The thesis: nothing is promoted on circular evidence

5

rescues PROMOTED

reclassified as reference-bias

3

rescues HELD by firewall

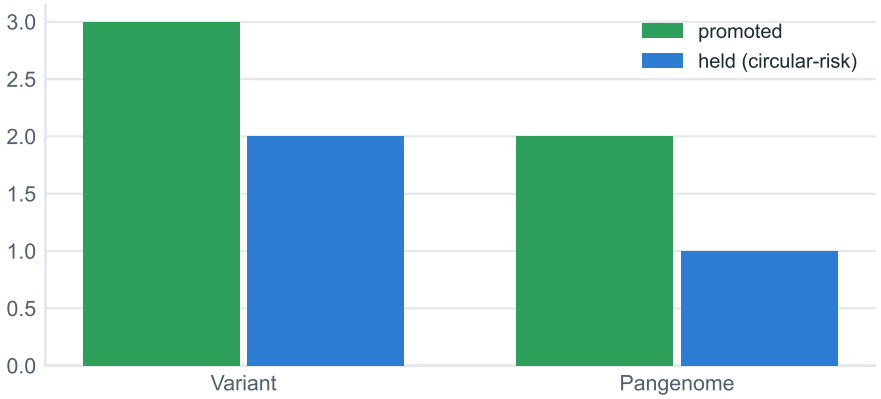
circular-risk provenance — not promoted

38%

firewall hold-rate

held / (held + promoted)

Rescue axis: promoted vs held



Held-rescue ledger (3)

isoform	category	why held (rule trace)
HELD_var.44	not_in_catalog	variant creates canonical motif on haplotype but prove
HELD_var.45	not_in_catalog	variant creates canonical motif on haplotype but prove
HELD_pan.46	not_in_catalog	all 2/2 novel junctions realizable on a pangenome path

Multi-caller consensus

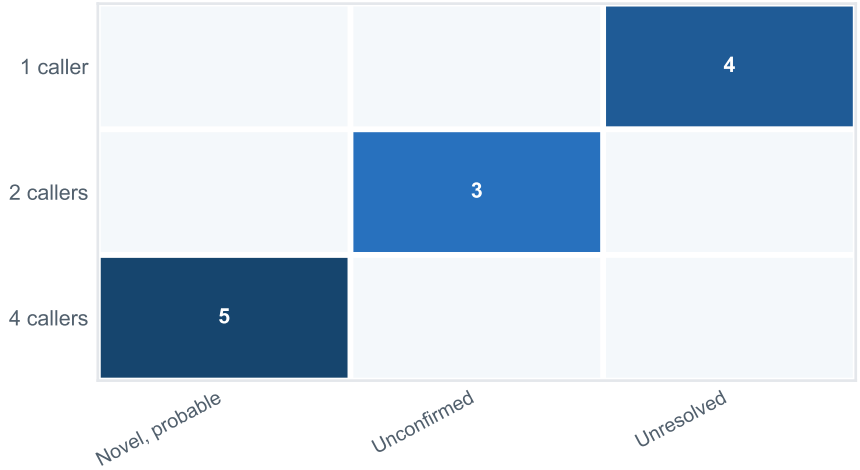
Caller agreement is methodological corroboration — it promotes to MEDIUM at most, never HIGH.

Caller-support distribution

how much do callers agree?



Caller support → confidence class



Verdicts of short-read-unknown novels under the consensus axis

promotions to MEDIUM / LOW come purely from cross-caller agreement

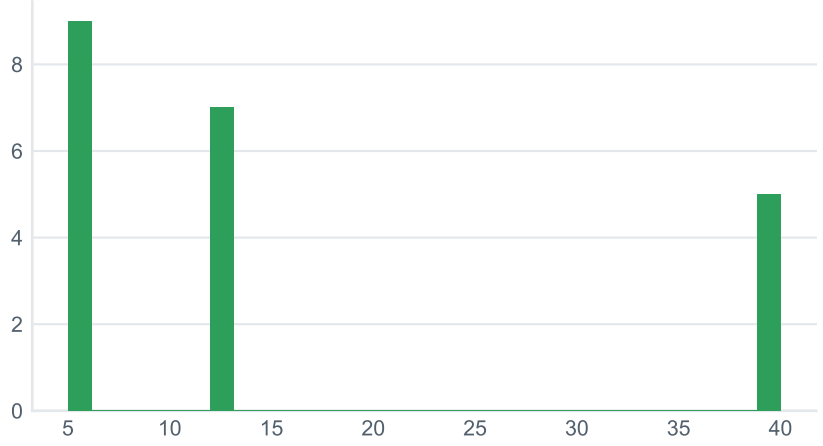


SQANTI3 biological priors

CAGE, polyA site and NMD are passed through and NOT used in the verdict; the polyA motif only exempts an A-rich 3' end from the intra-priming sign.

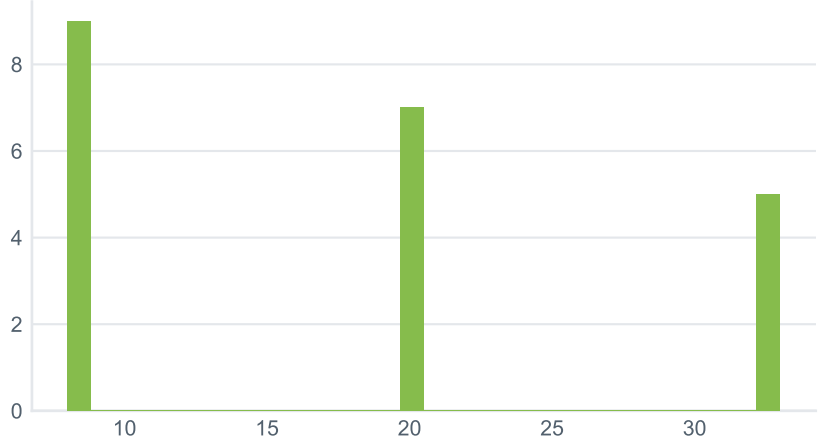
Distance to CAGE peak

clipped to ±500 bp



Distance to polyA site

clipped to ±500 bp



Predicted NMD



PolyA motif found

—

no polyA-motif prior

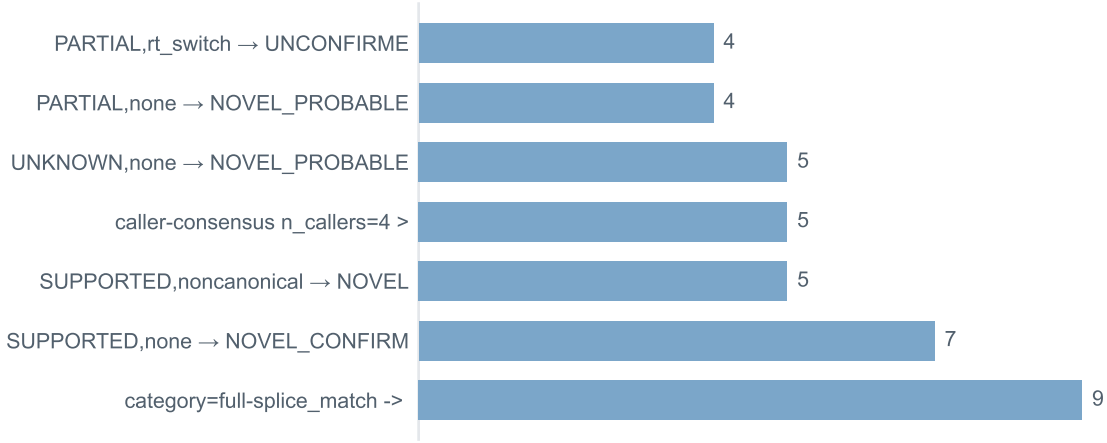
Provenance & defensibility appendix

Exactly how this report was produced, and the checks that make it auditable.

Run provenance

key	value
tool_version	0.0.6
ruleset_version	builtin-0.0.3
sqanti3_version_targe	6.0
config	<built-in defaults>
classification	showcase (synthetic demo)
isoforms	showcase (synthetic demo)
ref_gtf	showcase (synthetic demo)
sj_tab	showcase (synthetic demo)
bam	showcase (synthetic demo)
caller_support	showcase (synthetic demo)
axis.short_read	on
axis.catalog	on
axis.bam	on
axis.variant	on
axis.pangenome	on
axis.consensus	on

Most frequent rule-trace lines



Defensibility checklist

- ✓ class tally reconciles (provenance ↔ JSONL)
- ✓ firewall held 3 circular-risk rescue(s)
- • consensus axis used this run
- • axes in play: short-read, BAM, variant, pangenome, consensus
- • SQANTI3 priors consumed, never recomputed